

qSOLIS

User Manual

Version 1.1.0

Solar visibility, irradiance, irradiation, and canopy exposure analysis for CloudCompare

Release binaries: CloudCompare v2.14.beta2 / Qt 6

Windows x64 | macOS arm64 (unsigned) | Ubuntu 24.04 x86_64

Released 2026-10-24

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1. Installation and release compatibility

qSOLIS release archives contain the qSOLIS plugin binary only. They do not contain a complete CloudCompare installation. Install CloudCompare first, then copy the qSOLIS plugin into the matching plugin directory.

Platform	Architecture	Release target
Windows	x86_64 / x64	CloudCompare v2.14.beta2
macOS	Apple Silicon / arm64	CloudCompare v2.14.beta2
Linux	x86_64	Ubuntu 24.04, CloudCompare v2.14.beta2

Important: use the package matching both your operating system and architecture. The v1.1.0 binaries are built and tested against CloudCompare v2.14.beta2.

1.1 Windows x64

Download:

```
qSOLIS-1.1.0-CloudCompare-v2.14.beta2-windows-x64-msvc.zip
```

The archive contains:

```
CloudCompare\plugins\QSOLIS_PLUGIN.dll
```

1. Install or extract the 64-bit CloudCompare v2.14.beta2 Windows build.
2. Close CloudCompare if it is running, then extract the qSOLIS ZIP.
3. Copy QSOLIS_PLUGIN.dll into <CloudCompare installation>\plugins\.
4. Start CloudCompare and confirm that SOLIS appears in the plugin menu.

1.2 macOS Apple Silicon (arm64)

Download:

```
qSOLIS-1.1.0-CloudCompare-v2.14.beta2-macos-arm64-clang-unsigned.zip
```

The archive contains:

```
CloudCompare/CloudCompare.app/Contents/PlugIns/ccPlugins/libQSOLIS_PLUGIN.dylib
```

1. Install CloudCompare v2.14.beta2 for Apple Silicon (arm64) and quit CloudCompare.
2. Extract the qSOLIS ZIP.
3. In Finder, Control-click CloudCompare.app and choose Show Package Contents.
4. Open Contents/PlugIns/ccPlugins/ and copy libQSOLIS_PLUGIN.dylib into that directory.
5. Start CloudCompare and confirm that SOLIS appears in the plugin menu.

macOS note: the v1.1.0 plugin is unsigned. macOS may show a security warning. Only allow it after verifying that the archive came from the official qSOLIS release.

1.3 Linux x86_64

Download:

```
qSOLIS-1.1.0-CloudCompare-v2.14.beta2-linux-x86_64-ubuntu24.04-gcc.zip
```

The archive contains:

```
lib/cloudcompare/plugins/libQSOLIS_PLUGIN.so
```

1. Install CloudCompare v2.14.beta2 and close it if running.
2. Extract the qSOLIS ZIP.
3. Copy libQSOLIS_PLUGIN.so into <CloudCompare installation prefix>/lib/cloudcompare/plugins/.
4. Start CloudCompare and confirm that SOLIS appears in the plugin menu.

Linux note: the release is built and tested on Ubuntu 24.04 x86_64. Other distributions may require a source build because of compiler, Qt, or system-library compatibility.

1.4 Verify the downloaded archive

Each release ZIP is accompanied by a .sha256 file. On Linux/macOS:

```
sha256sum -c <package>.sha256
```

On Windows PowerShell:

```
Get-FileHash <package>.zip -Algorithm SHA256
```

2. Overview

qSOLIS evaluates solar exposure on CloudCompare point clouds and triangular meshes. It combines a solar-position/clear-sky irradiance model with an offscreen OpenGL visibility pass. Results are written as CloudCompare scalar fields on the original point or vertex cloud.

- Direct solar visibility or irradiance.
- Diffuse sky visibility or irradiance.
- Total irradiance/irradiation as Direct + Diffuse.
- Instantaneous irradiance in W/m2.
- Mean irradiance over an interval in W/m2.
- Integrated irradiation in Wh/m2 or kWh/m2.
- Horizontal-reference, true surface (normal-aware), and visibility modes.
- Canopy exposure metrics: SEF, SuEF, REI, and CEI.
- GUI and command-line workflows using the same numerical engine.

3. Coordinate and time conventions

qSOLIS assumes +X = east, +Y = north, and +Z = up. Longitude is east-positive; latitude is positive north and negative south.

Day-of-year (DOY) runs from 1 through 366. DOY 0 is invalid. A clock time is stored internally as the fractional part of the day-of-year.

Time can be entered as UTC or as local civil time with a fixed UTC offset. Local-time mode is a fixed offset only; it does not apply named timezone or daylight-saving rules automatically.

4. Starting a calculation

1. Load a point cloud or mesh.
2. Select the point cloud itself or the mesh in the DB tree.
3. Choose Plugins > SOLIS Calculate....
4. Configure the calculation.
5. Press OK.

The dialog displays qSOLIS v1.1.0 at the bottom so screenshots and reports can identify the plugin version.

4.1 Location and time

Set latitude, longitude, elevation, DOY, time, and time reference. For UTC, leave the UTC offset disabled. For local fixed-offset time, choose Local and enter the offset from UTC.

4.2 Output quantity

Mode	Output	Meaning
Instantaneous	W/m2	One time point
Mean	W/m2	Average over configured interval
Integrated energy	Wh/m2 or kWh/m2	Time-integrated irradiation

For Mean and Energy modes, Duration and Timestep become active.

4.3 Spatial mode

Mode	Use
Horizontal reference	Visible points receive horizontal-plane incidence weighting; target normals are not required.
True surface	Uses each point/vertex normal; normals are required.
Visibility	Returns visibility fractions and ignores irradiance magnitude.

If any selected point cloud or mesh vertex cloud lacks normals, Surface mode is disabled. In Visibility mode, Total is undefined; Direct + Diffuse creates two separate visibility fields.

4.4 Components

Select Direct, Diffuse, or both. When both are selected in an irradiance mode, qSOLIS also creates Total = Direct + Diffuse point-by-point.

4.5 Quality presets

Preset	Render resolution	Diffuse rays
Fast	512	64
Normal	1024	256
High	2048	512
Custom	User selected	User selected

Higher resolution improves shadow-boundary detail but increases GPU memory, framebuffer readback, and CPU projection work. Normal is the recommended starting point.

5. Advanced options

Expand Advanced options for controls that are normally left at defaults.

- Linke turbidity: clear-sky atmospheric attenuation; one constant value per calculation.
- Resolution: square visibility framebuffer size.
- Depth bias: depth-comparison tolerance used by the visibility test.
- CPU threads: Auto, 1 for single-threaded projection, or an explicit worker count.
- Closed mesh optimization: enable only for known watertight meshes.
- Diffuse rays: deterministic hemisphere sample count.
- Direct angular grouping: optional approximation that merges nearby sun directions for long integrations.

6. Batch time series

Enable Batch time series to store multiple calculation start times as separate scalar fields without duplicating geometry. Set an end DOY/time and a step in minutes. For Mean or Energy modes, each batch timestamp is the start of its own interval.

The GUI asks for confirmation for large batches and qSOLIS limits one batch to 1000 time points.

7. Result layers and cleanup

Each persistent result is a new scalar field on the original point/vertex cloud. Existing qSOLIS results are never silently overwritten.

```
SOLIS Direct [W/m2] D175 T1200 UTC H R1024
SOLIS Diffuse [W/m2] D175 T1200 UTC H R1024
SOLIS Total [Wh/m2] D175 T0000 UTC dur24h dt5m H R1024
SOLIS Direct Mean [W/m2] D175 T1200 UTC dur2h dt5m S R2048
```

Duplicate names receive #2, #3, and so on. Full calculation provenance is stored as qSOLIS JSON metadata, including plugin version, run/batch identifiers, location/time, atmosphere, rendering, and sampling settings.

Cleanup: select one or more clouds/meshes and choose Plugins > SOLIS Cleanup.... Normal cleanup is metadata-driven and leaves unrelated scalar fields untouched. Available operations include Last run, Last batch, Selected batch, All metadata-backed qSOLIS results, and explicit legacy SOLIS-name cleanup.

8. Command-line use

qSOLIS registers -SOLIS and -SOLIS_CLEAN. The complete option reference is maintained in the repository CLI documentation.

```
CloudCompare -SILENT -AUTO_SAVE OFF -O tree.ply -SOLIS \
  -COMPONENT DIRECT -OUTPUT_MODE INSTANT -SPATIAL_MODE HORIZONTAL \
  -DOY 172 -TIME 12:00 -LAT 45 -LON 0 -ELV 0 -RESOLUTION 1024
```

Every created field is printed to the CLI log. Automation should capture these exact names rather than reconstructing duplicate suffixes.

9. Canopy Exposure

Use Plugins > SOLIS Exposure... for obstruction-aware canopy and crown exposure metrics. These metrics are independent of target normals and are designed to quantify sky openness, direct-sun access, and radiative exposure.

9.1 Exposure metrics

Metric	Range	Meaning
SEF - Sky Exposure Factor	0-1	Weighted geometric visibility of the upper hemisphere. 1 is open sky; 0 is fully obstructed.
SuEF - Sun Exposure Fraction	0-1	Fraction of sampled daylight time with line of sight to the sun. Time-weighted, not DNI-weighted.
REI - Radiative Exposure Index	0-1	Received clear-sky direct + diffuse radiation divided by an unobstructed horizontal reference receiver.
CEI - Crown Exposure Index	0-1	Entity-level spatially weighted mean REI for a crown/tree target.

CEI note: qSOLIS CEI is a radiative exposure summary and is not the Dawkins ordinal Crown Illumination Index.

9.2 Occlusion context

The Occlusion context control determines which geometry can block rays. Only the target entity receives output scalar fields; context entities act as blockers.

Context mode	Behavior
Target only	Only the target geometry can occlude itself.
Selected entities	Selected neighboring entities also block rays. This is the default/recommended multi-entity workflow.
All visible entities	All visible clouds/meshes can act as occluders while results remain on the selected target.

For point-cloud occluders, apparent blocking depends on point density and render resolution. Solid meshes generally provide more continuous occlusion for walls/buildings when available.

9.3 Crown Exposure summary

CEI is computed from per-point/per-vertex REI using spatial weighting. Point clouds use equal occupied-voxel weights (averaging within each occupied voxel); meshes use triangle-area weights distributed to vertices. The default point-cloud voxel size is derived from the bounding-box diagonal and can be overridden.

Persisted crown metadata includes weighted mean SEF/SuEF/REI when available, REI P10/P50/P90, exposed fraction above the configured threshold (default 0.75), the weighting method, voxel size/count or represented surface area, occlusion context, and provenance.

9.4 Exposure command-line example

```
CloudCompare -SILENT -AUTO_SAVE OFF -O crown.ply \
  -SOLIS_EXPOSURE -METRICS SKY \
  -SKY_RAYS 256 -RESOLUTION 1024
```

For multi-entity CLI runs, -CONTEXT TARGET uses self-only occlusion and -CONTEXT LOADED allows separately loaded geometry to block the target.

10. Exporting one Crown Exposure row per tree

After calculating Crown Exposure, qSOLIS can convert persistent per-entity summaries into a CSV table. Keep each biological tree as a separate point-cloud/mesh entity, select the trees to export, and choose Plugins > SOLIS Export Crown Exposure Summary....

The GUI writes one CSV row per selected tree using its latest stored crown run. Missing optional values are left empty rather than written as zero.

The CSV includes CEI, Sky/Sun means when calculated, REI mean/P10/P50/P90, exposed fraction, voxel or mesh weighting information, occlusion context and occluder names, site/time settings, and run/plugin/schema provenance.

```
CloudCompare -O result.bin -SOLIS_EXPORT_CROWN crown_exposure.csv LATEST
CloudCompare -O result.bin -SOLIS_EXPORT_CROWN crown_exposure.csv ALL_RUNS
```

11. Interpretation of results

11.1 Direct, Diffuse, Total, Visibility

- Direct: clear-sky direct normal irradiance, visibility, and the selected incidence term. Occluded points receive zero direct contribution for that sample.
- Diffuse: isotropic sky radiance sampled over a deterministic hemisphere; Surface mode applies normal incidence.
- Total: Direct + Diffuse for irradiance/energy modes.
- Direct visibility over an interval: fraction of sampled daylight time visible to the sun.
- Diffuse visibility: fraction of sampled sky solid angle visible from the point/vertex.

11.2 Canopy metrics

- SEF near 1 indicates an open upper hemisphere; lower values indicate geometric obstruction.
- SuEF near 1 indicates direct sun access for most of the sampled daylight interval.
- REI near 1 indicates radiative exposure close to an unobstructed horizontal reference.
- CEI summarizes REI across a whole crown/entity using spatial rather than raw point-count weighting.

12. Performance guidance

1. Start at Normal quality (1024 / 256 rays).
2. Compare 1024 and 2048 before adopting High for production.
3. Use a coarser temporal timestep when scientifically acceptable.
4. Benchmark direct angular grouping against an ungrouped reference before relying on it.
5. Keep CPU threads on Auto unless benchmarking indicates otherwise.
6. For point-cloud occlusion tests, ensure point spacing is dense enough relative to the chosen render resolution.

qSOLIS reuses uploaded geometry and the rendering backend across Direct/Diffuse and batch points.

13. Regression tests

Release validation uses the installed CloudCompare executable, not a build-tree executable.

13.1 Frozen solar regression

```
cd qSolis/tests/regression
./run_qsolis_regression.sh /path/to/CloudCompare
```

Frozen result	Expected value
Surface horizontal median	262.5244 W/m2
Surface south median	665.6770 W/m2
Surface north median	0.0000 W/m2
45 deg south/up median	656.6156 W/m2
Wall sunlit median	834.6073 W/m2
Wall shadow length	1.9746 m

13.2 v1.1 exposure regression

```
./run_qsolis_exposure_regression.sh /path/to/CloudCompare
```

The v1.1 suite covers open-reference Sky/Sun/Radiative exposure, no-daylight rejection, wall Sun/Sky/REI behavior, and separate target/neighbor geometry for target-only versus loaded-context occlusion.

Context acceptance case	Expected behavior
Target only	Near = 1.0000, far = 1.0000 (neighbor ignored)
Loaded context	Near = 0.0000, far = 1.0000 (neighbor blocks target)

14. Troubleshooting

Symptom	Check
SOLIS action disabled	Select the actual point cloud or mesh, not only a parent container.
Surface mode disabled	Compute/import normals for every selected point/vertex cloud.
All-zero direct result	Verify sun above horizon and check location/time; inspect OpenGL errors if resolution-specific.
CLI on headless host fails	-SILENT removes dialogs but not the Qt/OpenGL requirement; use an offscreen-capable platform or virtual display.
Canopy neighbor does not occlude	Check occlusion context, entity visibility, point density, and render resolution; use a mesh for solid walls when available.
Too many result fields	Use SOLIS Cleanup > Last batch or another metadata-driven cleanup option.
macOS plugin blocked	The release is unsigned. Verify the release source and allow it through macOS security controls.

15. Scientific limitations

- No measured cloud cover or meteorological time-series input.
- Isotropic diffuse-sky model.
- Constant Linke turbidity over one calculation.
- Surface results depend on normal quality and orientation.
- Shadow boundaries depend on finite render resolution and depth bias.
- Point-cloud occlusion depends on sampling density relative to render resolution.
- Day-of-year solar approximations do not carry an explicit calendar year.

Document these assumptions when qSOLIS results are used in research or engineering reports.

16. Version and license

This manual describes qSOLIS v1.1.0, released 2026-10-24. Release binaries are built and tested against CloudCompare v2.14.beta2 / Qt 6 for Windows x64, macOS Apple Silicon arm64 (unsigned), and Ubuntu 24.04 x86_64.

qSOLIS is distributed under the GNU General Public License, version 2 or later. See the release LICENSE file.